

0860723 Selected Subjects In Robotics and AI

HW1: Estimation Signal and Image Processing

Question 1. Fourier-Based Signal Processing:

Consider the continuous-time signal

$$x(t) = e^{-a|t|} \cos(2\pi f_0 t), \text{ with } a > 0.$$

- (a) Derive analytically the Fourier Transform $X(f)$ of $x(t)$, showing all steps, and sketch the resulting magnitude spectrum.
- (b) Explain how the Fourier Transform can be used to separate signal components from additive noise.
- (c) Consider sampling $x(t)$ with sampling frequency f_s . Explain the phenomenon of aliasing and analytically show what happens in the frequency domain when $f_s < 2f_{max}$ here f_{max} the highest frequency present in the signal.
- (d) Discuss practical consequences of aliasing in signal and image processing systems and methods to prevent it.

Question 2. Convolutional Neural Networks (CNNs):

Consider a CNN composed of the following layers:

Input image of size $64 \times 64 \times 3$, a convolutional layer with 16 filters of size 3×3 , stride 1, followed by a ReLU activation, a 2×2 max-pooling layer, and a fully connected layer with 100 neurons leading to a 10-class softmax output.

- (a) Describe the role of each layer in the architecture.
- (b) Compute the spatial dimensions after each layer.
- (c) Calculate the total number of trainable parameters in the network.

Question 3. Anomaly Detection in Signals:

- (a) Explain how classical statistical methods (e.g., thresholding, spectral analysis, or Kalman filtering residuals) can be used for anomaly detection in time-series signals.
- (b) Describe encoder–decoder based methods such as autoencoders for anomaly detection, including the role of reconstruction error.
- (c) Compare the strengths and limitations of classical versus deep learning–based approaches.

Question 4. Visual Object Detection Using CNNs (Programming Question):

Implement a convolutional neural network for visual object detection using a public dataset (e.g., CIFAR-10, Pascal VOC, or a simplified subset of COCO).

- (a) Design and implement the CNN architecture.
- (b) Train the model and evaluate its detection or classification performance using appropriate metrics.
- (c) Analyze the influence of key hyperparameters such as learning rate, number of convolutional layers, filter size, and batch size.
- (d) Present qualitative and quantitative results, including loss curves, accuracy or mAP, and example detections. Discuss your findings.